Data Literacy and Research Data Management in Two Top Universities in Poland. Raising Awareness

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Abstract. We present the results of the Polish part of a study conducted within the framework of the international research project named ReDaM coordinated by the Information Literacy Association (InLitAs). It was a quantitative study based on use of a questionnaire consisted of 25 open-ended and multiple choice questions that was translated from English into Polish. Data were collected from February to the end of April 2017. The target groups were doctoral students and faculty employed at the University of Warsaw and Wrocław University of Science and Technology. The study revealed that a significant number of respondents knew the basic concepts related to Research Data Management (RDM). At the same time, they had not used institutional solutions elaborated in their parent institutions. We did not notice any differences in RDM practices between the fields of study at the two universities. However, we did notice significant differences between academic staff and research students.

Keywords: Data literacy · Data information literacy · Science data literacy
Research Data Management · Scholarly information literacy · Poland

1 Introduction

There are several reasons why Research Data Management (RDM) has become an emerging topic of the 21st century discussion among librarians and other information professionals. The common use of ICT tools made research more intensive, technology- and data-driven and led to handling huge volumes of data. However, still most researchers have not been formally trained to manage their own data [1–3]. In this field, RDM has become an area of keen interest in higher education, leading to considerable investment in services, resources, and infrastructure to support researchers' data management needs [4].

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2 Research Data Management–Labelling the Concepts

Terminology confusions happen often while talking about new concepts that have not yet gained global recognition. Here we provide a brief revision of existing terms with an attempt to put them in order to avoid terminology confusion. Also, for the sake of epistemology, it is valuable to investigate the origin and nature of RDM.

Since the education in the field of RDM is strongly connected with educational services offered by libraries, it may be considered as deriving from information literacy (IL), which in turn is linked to the concept of data information literacy. The foregoing term is closely connected to research data services that include RDM, according to Koltay [5] or science data literacy (SDL). In their work, Qin and D'Ignazio [6] stated that, although SDL is somehow similar to information literacy and digital literacy, it focuses less on literature-based attributes and more on functional ability in data collection, processing, evaluation, and use. The concept of digital literacy was described by the American Library Association (ALA) as the ability to use information and communication technology to find, evaluate, create, and communicate information [3].

Research data, as defined by Horizon 2020, the EU Framework Programme for Research and Innovation [7] refers to information, in particular facts or numbers, collected to be examined and considered and as a basis for reasoning, discussion or calculation. The Association of College and Research Libraries (ACRL), for the purpose of its 2014 survey among library professionals, defined RDM as the activities necessary for documenting, maintaining, and making the data collected in the course of research accessible. The data are discipline agnostic and may be in any format [2]. Makani [8] considers RDM in the context of data and information and knowledge discourse; she arguments that effective RDM is one that contributes to innovative and diverse knowledge discoveries, knowledge use, and knowledge reuse.

3 Research Data Management Practices

3.1 Researchers' Awareness

According to literature, there is a little awareness and low levels of comfort and expert self-assessment with the life cycle of research data and RDM topics. This opinion was confirmed by several studies (see e.g. [2, 9, 10]). Moreover, there is a phenomenon that can be named "personalization" or "individual habits"—we can observe individual habits and rules of each organization, research unit, or individual researcher. There is a lack of common practice, the research cultures differ, and a great resistance to changes may be observed [11]. As Burgi et al. [9] remarked, it is difficult to harmonize any of these habits. An OCLC Research Report underlines the importance of education services in raising RDM awareness and it points at two main services: expertise and curation [4]. Libraries have already years of experience in both of these services.

3.2 Identifying and Analyzing the Gaps

Planning initiatives aiming for development of RDM on national level requires an in-depth exploration of existing solutions in this field to avoid duplication and to identify the gaps. The Swiss project may be a good example of such an undertaking [9]. Also, the ACRL Digital Curation Interest Group conducted in 2014 a survey on professional development needs, aimed at librarians involved in the management and curation of research data at their institutions [2].

3.3 Data Management Plan (DMP)

A good DMP helps researchers describe what they are going to do with data during and after their research project, to set up a level of data openness (and thus, possibility of sharing) or confidentiality, and to define the period and conditions of data preservation after the research is completed. DMPs are important part of the research data cycle and they are more and more often required by funding bodies and research institutions [1, 9, 12, 13].

3.4 Existing RDM Education in Different Countries

Not surprisingly, US academic librarians lead in RDM education. Massachusetts Institute of Technology, the University of California, Berkeley, the University of Wisconsin-Madison, Purdue University, Syracuse University, and the University of Minnesota are just few examples [1, 2, 6, 14, 15]. ACRL offers the support of professional development in the field of RDM for librarians. In Europe, under the already mentioned national Swiss project, a group of librarians has been already trained by computer engineers and information specialists and, a website with relevant resources, tools, and guidelines has been created [9]. The University of Tartu, Estonia has offered an RDM course for doctoral students since 2015. The training is integrated into curricula. It is based on personal needs of young researchers and offers individual tutoring [16]. The University of Bristol, UK is developing RDM services; the project focuses on data management assistance, RDM advocacy, and training for researchers representing all disciplines [17].

3.5 Main Conclusions of the RDM Literature Review

RDM depends on institutional strategies and research habits in specific disciplines [9]. RDM education is in the scope of expertise and interest of librarians, however, it should not be provided only by this one group but with close cooperation with other specialists. Pre-assessment tests are crucial before planning and establishing any RDM educational initiatives. One of the ways of such a pre-assessment and a tool to raise awareness of RDM may be a pilot study [6].

4 Polish Study

The study conducted in Poland was a part of international research project, ReDaM, coordinated by the Information Literacy Association (InLitAs) that aimed to collect data about data literacy of academics and research students in higher education institutions. In this paper we present selected results. The purpose of the study was to explore the ways of research data management (RDM) by academic staff and research students and to assess the RDM awareness level of both target groups. Particularly, we wanted to check: (1) what are the Research Data Management practices? (2) are there differences in Research Data Management practices between academic staff and research students? (3) are there differences in Research Data Management practices between two types of higher education institutions, general university vs. university of technology?

4.1 Method

It was a quantitative study. The online survey on the LimeSurvey platform was established to collect data. The questionnaire consisted of 25 open-ended and multiple choice questions, translated from English into Polish. Data were collected from February to the end of April 2017. In this paper, we present only a part of the study, thus, we analyzed responses to selected questions.

4.2 Field

The target groups were doctoral students and faculty employed at the Wrocław University of Science and Technology (WUST) and the University of Warsaw (UW). These two large public universities are highly-ranked, holding second and first positions respectively, in the national rankings of technical universities and universities in Poland, Perspektywy 2016 [18]. These sites were chosen for this research to cover the largest possible number of fields to be represented in the study.

WUST is one of the most dynamically developing technical universities in the country. In 2016 it was granted the logo, "Human Resources Excellence in Research", by the European Commission. This logo is given to institutions that apply the principles of the European Charter for Researchers and the Code of Conduct for the Recruitment of Researchers. There are approximately 34,000 students (including over 800 doctoral students) and over 2,000 research and teaching staff at WUST. WUST has 5,452 registered inventions, over 12,000 publications in journals on the "Clarivate Analytics" Master Journal List, and over 11,000 publications in JCRI indexed journals. What is particularly interesting, and what makes difference between traditional university and the technical one, is that research in WUST focuses on the demands of the industry [19]. Over 74% of participants of the survey conducted during the implementation process of The European Charter for Researchers agreed that WUST provided researcher, including disabled researchers, with flexible working conditions in order to achieve effective research results. And over 60% claimed that WUST supported scientists at all stages of their careers, helping them develop and improve their skills and qualifications and enabling participation in conferences, trainings, and other forms of education [20].

UW is the largest higher education institution in Poland. It is composed of 30 academic and research units and 21 faculties. There are approximately 44,400 students and 6,200 doctoral and postgraduate students (studying, respectively, at 35 and 132 programs). Ten thousand students graduate per year. UW has 3,750 academic teachers, 785 of whom are full professors. It is a research-intensive university: it conducts currently over 1,500 projects financed by national or international research programs, funds, and organizations and produces about 8,700 scientific publications per year. The research potential of UW reflects in the results of the 2016 Nature Index of Rising Stars. The University of Warsaw was ranked 3rd among the institutions of South East Europe. Globally it occupied 96th place [21].

4.3 Data Analysis

We reported descriptive statistics including characteristics of the practices of research data use. The χ^2 test was used to examine differences between two samples. The χ^2 test was used to examine relationships between: (1) some practices and current primary roles, (2) some practices and types of higher education institutions.

Respondents' Characteristics. At WUST, the questionnaire was sent to 5266 respondents and was disseminated via the library webpage. At UW, library authorities denied use of the library mailing list to disseminate the questionnaire; the link to the survey was disseminated through the library webpage and social media platforms. The description and main goals of the study were described precisely in a short memo distributed on every distribution channel. It was filled out by 128 respondents; this low response rate a significant limitation in terms of generalization of results. The low number of respondents may be due to a relatively low number of concerned research students in Poland toward open data issues as well due to little awareness of the life cycle of research data and RDM topics among researchers as proved by several studies mentioned earlier in this paper. That is why the results presented in this paper should be treated as preliminary and refer to the sample only rather than to the whole population. There were more respondents from WUST: 83 (64.8%) than from UW: 45 (35.2%). Among the respondents there were: academic staff (n = 65, 50.8%), and research students (n = 58, 45.3%) and people who indicated other current primary roles (n = 7, 3.9%). The number of academic staff and research students did not differ significantly. The majority of respondents were below the age of 35 years (n = 82, 64.6%), fewer were between 36–45 years (n = 26, 20.5%), and the smallest group above 45 years (n = 19, 14.9%). The gender distribution did not vary significantly; there were 73 (57%) male and 54 (42.2%) female respondents. The largest number of respondents represented engineering and technology (n = 52, 40.6%), natural sciences (n = 28, 21.9%), social sciences (n = 26, 20.3%), and humanities (n = 20, 15.6%) and the smallest, medical and health sciences (n = 2, 1.6%). The low number of medical and health science respondents is explained by the fact that in Poland this is the domain of medical universities and not general or technical higher education institutions. The majority of respondents indicated Poland as their country of origin (n = 126, 98.4%) only two respondents declared another country (The Netherlands, Ukraine).

Practices of Research Data Use. Table 1 presents responses to the questions related to Research Data Management.

Table 1. Distribution of responses to the questions related to Research Data Management

Awareness and application of Research Data Management	Yes		No		Uncertain		χ^2
	n	%	*n*	%	*n*	%	
Familiar with the term 'metadata'	92	71.9	25	19.5	11	8.6	87.86***
Use some standard style for citing research data	76	59.4	43	33.6	9	7	52.61***
Familiar with the concept of Digital Object Identifier (DOI)	82	64.1	40	31.3	6	4.7	67.94***
Have got some unique research identification (for ex. ORCID)	37	28.9	77	60.2	14	10.9	47.64***

Note. *$p < .05$, **$p < .01$, ***$p < .001$.

As Table 1 illustrates, the majority of respondents knew the term 'metadata' (71.9%). A significant number (59.4%) of respondents used a particular standard style for citing research data than did not (33.6%). There were also significant differences in the awareness of what DOI is with 64.1% reporting they were aware of this identifier and 31.3% not aware. A significant difference also was seen in the number of respondents who had (28.9%) a unique research identification (for ex. ORCID) when compared with those who did not have (60.2%) a unique research identification. As indicated by the χ^2 test, there were no statistically significant differences in the answers on the questions related to RDM—presented in Table 1—between the respondents from UW and WUST ($\chi^2 = 3.79$, $df = 2$; $p > .05$, $\chi^2 = 1.75$, $df = 2$; $p > .05$, $\chi^2 = 3.74$, $df = 2$; $p > .05$, respectively).

The χ^2 test also showed that primary role of respondent (academic staff, research student) was not connected with the familiarity with the term 'metadata' ($\chi^2 = .1$, $df = 2$; $p > .05$). We also did not notice any differences between these two groups regarding their familiarity with the concept of Digital Object Identifier ($\chi^2 = 3.74$, $df = 2$; $p > .05$). This result could be explained by the fact that the use of this identifier as a searching tool is gaining popularity in academic environment. Significantly more academic staff (40%) had some unique researcher identification when compared to researcher students (17.2%) ($\chi^2 = 8.68$, $df = 2$; $p < .05$).

Research Data Management Plan. Table 2 illustrates answers to the questions related to RDMP.

As is shown in Table 2, the majority of respondents answered 'No' or 'Uncertain' to various questions related to RDMP. This indicates that respondents have not applied RDMP in their research projects. They also did not know institutional RDMP policies that might be useful in planning individual solutions (only three respondents knew institutional RDMP). However, more respondents recognized (19.5%) rather than did not recognize (13.3%) the importance of RDMP as a supporting tool for researchers in

Table 2. Distribution of responses to the questions related to Research Data Management Plan

Awareness and application of Research Data Management Plan (RDMP)	Yes		No		Uncertain		χ^2
	n	%	*n*	%	*n*	%	
Institution has RDMP	3	2.3	23	18	102	79.7	133.42***
Has ever used RDMP for her/his research	7	5.5	104	81.3	17	13.3	133.42***
Has RDMP for her/his current research project	8	6.3	106	82.8	14	10.9	141.44***
Is of opinion that RDMP helps researchers in managing research data	25	19.5	17	13.3	86	67.2	66.77***
Is of opinion that formal training on metadata would be useful for managing research data	73	57	13	10.2	42	32.8	42.2***

Note. *$p < .05$, **$p < .01$, ***$p < .001$.

their work with raw data. More (57%) noticed the usefulness of formal training for managing research data than did not notice (10.2%).

As the χ^2 test indicated, there were no statistically significant differences in answers to the questions related to RDMP—presented in Table 2—between the respondents from UW and WUST ($\chi^2 = 2.29$, $df = 2$; $p > .05$, $\chi^2 = .14$, $df = 2$; $p > .05$, $\chi^2 = 2.19$, $df = 2$; $p > .05$, $\chi^2 = 2.29$, $df = 2$; $p > .05$, $\chi^2 = 3.79$, $df = 2$; $p > .05$, respectively). Thus, the awareness and application of RDMP did not depend on the type of university.

The χ^2 test also revealed that the primary role of respondent (academic staff, research student) was not connected to the answer whether the institution had RDMP ($\chi^2 = 2.29$, $df = 2$; $p > .05$), whether the respondents had ever applied RDMP ($\chi^2 = 1.5$, $df = 2$; $p > .05$), or whether she or he used RDMP in the current project ($\chi^2 = 3.01$, $df = 2$; $p > .05$). In turn, academic staff and research students differed in their opinion on the role of RDMP as a tool that helps researchers in managing research data ($\chi^2 = 8.34$, $df = 2$; $p < .05$). Academic staff are considerably more often convinced that RDMP supports researchers in use of raw data (24.6%) than research students (12.1%). However, research students (65.5%) were considerably more often convinced—compared to academic staff (46.2%)—about the usefulness of formal training on metadata. The explanation of this results may be correlated in the general, quite specific attitude of Polish researchers towards institutional information literacy.

5 Discussion

The study revealed that a significant number of respondents knew the basic concepts related to Research Data Management. At the same time, they had not used institutional solutions, including the Research Data Management Plan in particular, provided by their parent institutions. Most frequently, the respondents were not aware of existence

of such solutions. We did not notice any differences in Research Data Management practices and knowledge of Research Data Management Plan between fields of study at the two universities. However, we did notice significant differences between academic staff and research students in their opinion on the usefulness of RDMP (academic staff more often than research students perceived this tool as useful) and on the usefulness of formal training on metadata (research students more often than academic staff perceived this kind of training as useful). In Research Data Management practice, academic staff differed from research students only in having a unique researcher identification. Academic staff significantly more often than research students had this identifier. This might be linked to their more extensive experience in scholarly publishing since more and more editors require ORCID from the submitting author.

6 Conclusion

Establishing RDMP helps researchers from the beginning of their projects in proper and active thinking about management of research data and its application. It helps avoid later doubts on data authorship and ownership. It is a first step that should be undertaken while planning research. This calls on researchers to consider which type of data will be produced, how much, where it will be stored and for how long preserved, and whether it will be shared or not. Last but not least, RDMP helps save time at the stage of data (or—more generally—research results) publishing. Formal, institutional training on RDMP seems to be the initial phase of data literacy education in Poland. Since June 2015, regular workshops on RDM are organized in Poland approximately twice a year by the Open Science Platform (Pl. *Platforma Otwartej Nauki* http://pon.edu.pl). Still, they are not popular and promoted enough to get to the public awareness of Polish academic community. The workshops, are one-day event for a limited number of participants. Since they are organized on-site in Warsaw they are not easy accessible for many researchers from other cities. That is why some other means of RDM awareness and promotion should be implemented. For example, in the Swiss project mentioned earlier in this paper, one of the expected results are guidelines for researchers and information professionals on RDM to encourage the use of best practices in data management [9].

However, as Carlson et al. [1, p. 635] stated, "any attempt to define a data information program must be aligned with current disciplinary practices and cultures if it is to be relevant to and accepted by its intended audience(s)". We believe that the Polish contribution into the collective intelligence of international ReDaM research project will help fulfill these requirements with regard to recommendations of LIBER working group on E-Science: "when exploring new services, make room for experiments/pilots, keep what works and learn from your mistakes" [22, p. 2].

It should also be mentioned that, according to Polish Law on Higher Education, universities have a statutory obligation to introduce regulations on intellectual property rights management. According to Article 86c of the Law on Higher Education [23], regulations for the management of copyright and related rights, and industrial property rights, as well as rules for the commercialization of research and development findings shall determine, among others: (1) rights and obligations of higher education institutions,

academic staff, students and doctoral students with reference to the use and protection of copyright and related rights as well as industrial property rights; (2) rules and procedures to govern the commercialization of research and development findings; (3) rules for the use of assets and facilities of a higher education institution for the purpose of commercialization of research and development findings, and for the provision of scientific and research services. Both WUST [24] and UW [25] have these kind of regulations. Thus, we can risk saying that there is a lack of coherence between introduced—at least at general level—related to regulations on RDM and everyday practice of academic community. We can also with high probability indicate this lack of coherence as an argument for the need of intensification of training on RDM for academic staff.

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